

Ali Abdallah

Junior AI Engineer

✉ eng.aliabdallah7@gmail.com |  [ali-abdallah7](#) |  [aliabdallah7](#) | ☎ +201126880776 | 📍 *Cairo, Egypt* | Military Service: **Completed**

Computer Science fresh graduate with 2+ years of project experience developing end-to-end machine learning and NLP solutions. Strong foundation in Object-Oriented Programming, Data Structures, Algorithms, Design Patterns, and Databases. Delivered multiple classification systems and deployed ML applications. Familiar with RAG and Agentic AI frameworks.

Education

B.Sc. in Computer Science, Ain Shams University (Class of 2024).

📅 October 2020 - July 2024

Department of **Information Systems**

📍 *Cairo, Egypt*

Grade: (B)

Graduation Project: Masha'erohom - Arabic Emotion Detector (**Grade: A+**)

Technical Skills

Programming: Python, R, C++, C#, Java, SQL (MySQL, SQLite), MongoDB, Git, Bash, FAISS, ChromaDB

Core CS: OOP, Data Structures, Algorithms, Design Patterns, System Design

Data & AI: Data Analysis, Machine Learning, Deep Learning (TensorFlow, PyTorch, Scikit-learn, Gradio), NLP, Data Visualization, Statistical Analysis, Statistical Inference, RAG, Agentic AI, LangChain, LlamaIndex

Selected Projects

Masha'erohom - Arabic Emotion Detector 

📅 August 2023 - July 2024

Graduation Project (Grade: A+)

- Led a team to develop an Arabic emotion detection model exceeding state-of-the-art accuracy (**92%**, **+7%** vs. prior work).
- Developed and architected an ensemble of **Bi-LSTM**, **Bi-GRU**, and **fine-tuned MARBERTv2**, stacked via **Random Forest** after extensive **hyperparameter tuning** and evaluation.
- Fine-tuned pretrained Arabic **word embeddings** to better capture dialectal nuances, diacritics, and morphology.
- Built a **Flask** web app that ingests Twitter data and visualizes public emotion distributions for user-entered keywords.
- Published a **peer-reviewed paper**  following formal evaluations in IJICIS.

English Sign Language Recognition 

📅 August 2023 - September 2023

- Developed a **CNN**-based model using landmark hand detection, achieving **99%** test accuracy.
- Integrated Random Forest for real-time gesture classification.

Spam Detection 

📅 April 2023 - April 2023

- Built an NLP-based spam classification system using **TF-IDF** and Count Vectorization to detect unwanted SMS messages.
- Trained and evaluated multiple classifiers (**Logistic Regression**, **SVM**, **KNN**, Random Forest, **Naive Bayes**) to select the best-performing model.
- Deployed the final model as an interactive **Streamlit** web application enabling real-time predictions.

Loan Eligibility Predictor

 February 2023 - February 2023

- Developed a machine learning model to automate loan eligibility prediction based on applicant financial data.
- Performed data preprocessing and feature engineering, then trained and compared multiple classification algorithms.
- Deployed the selected **Logistic Regression** model as a user-facing **Streamlit** web application for instant eligibility assessment.

Customer Segmentation

 November 2022 - December 2022

- Performed unsupervised learning (**K-Means**, **DBSCAN**, **Hierarchical**) using R for market segmentation.
- Conducted EDA and visualized insights to support business strategy.

Relevant Experience

Machine Learning & Computer Vision Trainee

 August 2023 - September 2023

ApplAi - Ain Shams University

- Completed intensive training in **Deep Learning and Computer Vision**, covering **CNN** fundamentals, modern architectures, object detection, and image segmentation.
- Built and evaluated multiple computer vision models using Python and deep learning frameworks.
- Achieved **1st place in 3 out of 5 Kaggle competitions**, demonstrating strong model optimization and problem-solving skills.
- Graduation project ranked in **Top 10** among all participants.

For more information about the competitions and the notebooks , the Completion Project  and the certificate .

Machine Learning Trainee

 February 2023

ApplAi - Ain Shams University

- Gained hands-on experience with supervised and **unsupervised machine learning algorithms**.
- Applied **end-to-end ML workflows** include data preprocessing, feature engineering, model training, and evaluation.
- Delivered a complete machine learning project as part of the program.

Here is the Project and the certificate .